

# Cell Requirements – Energy, Nutrients, and Waste Removal

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Every living cell is a tiny factory. It never stops working. It needs raw materials (nutrients). It needs power (energy). And it must take out the trash (waste removal). From the moment a cell is born until it divides or dies, it constantly exchanges matter and energy with its surroundings. Yet most people never consider what a cell needs to stay alive — or how plants have solved the energy problem in a completely different way than animals.

Biology is not merely a subject taught in classrooms. It is the study of how life sustains itself. From the smallest bacterium to the largest redwood tree, from the neurons in your brain to the muscle cells in your heart, every cell requires energy, nutrients, and waste removal. Understanding cell requirements means understanding why you need to eat, breathe, and eliminate waste — and how plants feed themselves (and the rest of the planet) using nothing but sunlight, water, and air.

This reading material explores the fundamental requirements of all cells, the process of photosynthesis in autotrophs, the role of chloroplasts, and the critical function of chlorophyll and other pigments in capturing light energy.

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## Part 1: What Do All Cells Require and Why Does It Matter?

### The Three Universal Requirements of All Cells

Every living cell — whether prokaryotic or eukaryotic, plant or animal, bacterial or human — has three non-negotiable requirements:

| Requirement          | Why Cells Need It                                                                                                   | What Happens If Missing                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Energy</b>        | To power chemical reactions, transport molecules, grow, and divide                                                  | Cell stops functioning; death within minutes to hours |
| <b>Nutrients</b>     | To provide raw materials for building cellular structures (membranes, proteins, DNA) and to serve as energy sources | Cell cannot grow or repair itself; eventually starves |
| <b>Waste removal</b> | To eliminate toxic byproducts of metabolism (carbon dioxide, ammonia, urea)                                         | Toxins accumulate; pH changes; cell poisoning occurs  |

## Why Cell Requirements Matter in Biology

| Aspect             | Explanation                                                        | Examples                                                               |
|--------------------|--------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Survival</b>    | Cells cannot live without continuous energy and nutrient supply    | Brain cells die after 3-5 minutes without oxygen (needed for energy)   |
| <b>Metabolism</b>  | All chemical reactions in cells depend on energy and raw materials | Building proteins requires energy (ATP) and amino acids (nutrients)    |
| <b>Homeostasis</b> | Waste removal keeps internal conditions stable                     | CO <sub>2</sub> buildup lowers blood pH (acidosis), which can be fatal |
| <b>Nutrition</b>   | Understanding cell requirements explains why organisms need food   | Humans need carbohydrates, fats, proteins, vitamins, minerals          |
| <b>Medicine</b>    | Many drugs target cell requirements to treat disease               | Chemotherapy targets rapidly dividing cancer cells' energy needs       |

## How Different Cells Obtain Energy and Nutrients

All cells fall into two categories based on how they obtain energy and nutrients:

### Autotrophs (Self-Feeders)

**Definition:** Organisms that produce their own food from inorganic sources

**Energy Source:** Light (photoautotrophs) or chemicals (chemoautotrophs)

**Carbon Source:** Carbon dioxide (CO<sub>2</sub>)

**Examples:** Plants, algae, cyanobacteria

### Heterotrophs (Other-Feeders)

**Definition:** Organisms that obtain energy and nutrients by consuming other organisms

**Energy Source:** Chemical bonds in organic molecules (food)

**Carbon Source:** Organic compounds (sugars, fats, proteins)

**Examples:** Animals, fungi, most bacteria

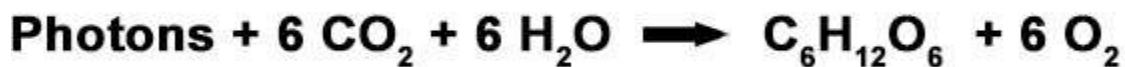
**Key insight for students:** Autotrophs are the **producers** of the biosphere. Heterotrophs are the **consumers**. Without autotrophs, heterotrophs would have no food or oxygen.

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## Part 2: Photosynthesis – How Autotrophs Capture Energy

**Photosynthesis** is the process by which autotrophs (plants, algae, and cyanobacteria) convert light energy from the sun into chemical energy stored in glucose (sugar). This glucose is then used by the cell for energy or built into other organic molecules.

The overall balanced chemical equation for photosynthesis is:



## Where Photosynthesis Occurs – The Chloroplast

Photosynthesis takes place within specialized organelles called **chloroplasts**. These are found only in the cells of autotrophs (plants and algae). Prokaryotic autotrophs (cyanobacteria) do not have chloroplasts; they have chlorophyll embedded in folded membranes.

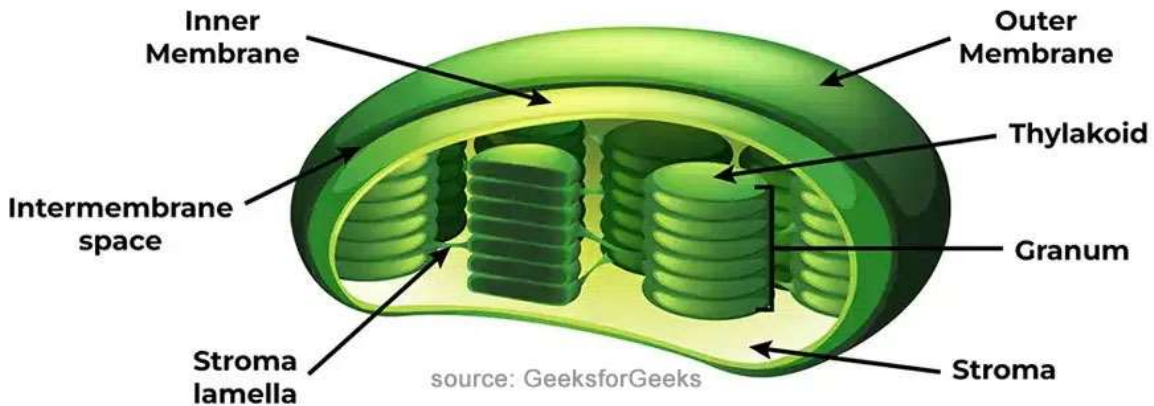
### Structure of a Chloroplast (Draw this)

| Part of Chloroplast             | Description                                           | Function                                                                           |
|---------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Thylakoid</b>                | Flattened, disk-shaped sacs arranged in stacks        | Light-dependent reactions occur here (chlorophyll embedded in thylakoid membranes) |
| <b>Grana</b> (singular: granum) | Stacks of thylakoids                                  | Increases surface area for light absorption                                        |
| <b>Stroma</b>                   | Fluid-filled space surrounding the thylakoids         | Light-independent reactions (Calvin cycle) occur here                              |
| <b>Chlorophyll</b>              | Green pigment molecule embedded in thylakoid membrane | Captures light energy (absorbs blue and red light; reflects green)                 |

## Chloroplasts Are Found Only in Autotrophs

| Organism Type                    | Contains Chloroplasts? | Where Photosynthesis Occurs                    |
|----------------------------------|------------------------|------------------------------------------------|
| Plants (leaves, green stems)     | Yes                    | Chloroplasts in mesophyll cells                |
| Algae                            | Yes                    | Chloroplasts                                   |
| Cyanobacteria (blue-green algae) | No (prokaryotes)       | Chlorophyll in folded thylakoid-like membranes |
| Animals                          | No                     | Cannot perform photosynthesis                  |
| Fungi                            | No                     | Cannot perform photosynthesis                  |

# Diagram of a Chloroplast



## Part 3: The Role of Chlorophyll and Other Pigments

**Chlorophyll** is the primary pigment that captures light energy for photosynthesis. However, plants contain several pigments that work together to maximize light absorption.

### Types of Photosynthetic Pigments

#### Chlorophyll a

**Color:** Blue-green

**Wavelengths Absorbed:** Red light (660-665 nm) and violet-blue light (430-450 nm)

**Wavelengths Reflected:** Green (this is why plants look green)

**Role:** Primary pigment; directly converts light energy to chemical energy

## Chlorophyll b

**Color:** Yellow-green

**Wavelengths Absorbed:** Blue and orange light

**Wavelengths Reflected:** Yellow-green

**Role:** Accessory pigment; captures light and transfers energy to chlorophyll a

## Carotenoids (beta-carotene, lutein)

**Color:** Yellow, orange, red

**Wavelengths Absorbed:** Blue-violet light

**Wavelengths Reflected:** Yellow, orange, red

**Role:** Accessory pigments; protect against excess light (photoprotection); give autumn leaves their colors

## How Pigments Capture Light Energy

| Step | What Happens                                                                    |
|------|---------------------------------------------------------------------------------|
| 1    | Light strikes the pigment molecule (chlorophyll or accessory pigment)           |
| 2    | The energy from light excites electrons in the pigment to a higher energy level |
| 3    | Excited electrons are transferred to electron acceptor molecules                |
| 4    | This energy is used to make ATP (energy currency) and NADPH (electron carrier)  |
| 5    | Accessory pigments pass their captured energy to chlorophyll a                  |

## Why Plants Have Multiple Pigments

Plants do not just have chlorophyll a. They have a team of pigments for two reasons:

| Reason                                    | Explanation                                                                                                                                                                                                                  |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capture more of the light spectrum</b> | Chlorophyll a absorbs mainly red and blue light. It does NOT absorb green or yellow light well. Carotenoids and chlorophyll b capture light in other wavelengths (blue-violet, orange) and pass the energy to chlorophyll a. |
| <b>Protection from excess light</b>       | Too much sunlight can damage chlorophyll and cell structures. Carotenoids absorb excess energy and release it as heat, protecting the cell from damage. This is called <b>photoprotection</b> .                              |

## Visible Light Spectrum and Pigment Absorption

| Color of Light | Wavelength | Absorbed by Which Pigment?                        |
|----------------|------------|---------------------------------------------------|
| Violet         | ~400 nm    | Chlorophyll a, carotenoids                        |
| Blue           | ~450 nm    | Chlorophyll a, chlorophyll b, carotenoids         |
| Green          | ~550 nm    | Poorly absorbed (reflected) — plants appear green |
| Yellow         | ~570 nm    | Chlorophyll b, carotenoids                        |
| Orange         | ~600 nm    | Chlorophyll b, carotenoids                        |
| Red            | ~660 nm    | Chlorophyll a                                     |

**Key point:** Plants are green NOT because they need green light, but because chlorophyll reflects green light while absorbing red and blue light.

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## Part 4: The Two Stages of Photosynthesis

Photosynthesis occurs in two main stages: the **light-dependent reactions** and the **light-independent reactions (Calvin cycle)** .

| Feature                        | Light-Dependent Reactions                       | Light-Independent Reactions                                                      |
|--------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Location in chloroplast</b> | Thylakoid membrane                              | Stroma (fluid)                                                                   |
| <b>Requires light?</b>         | Yes — light required                            | No — can occur in dark (but needs products of light reactions)                   |
| <b>Inputs</b>                  | Light, H <sub>2</sub> O, ADP, NADP <sup>+</sup> | CO <sub>2</sub> , ATP, NADPH                                                     |
| <b>Outputs</b>                 | O <sub>2</sub> (oxygen), ATP, NADPH             | Glucose (C <sub>6</sub> H <sub>12</sub> O <sub>6</sub> ), ADP, NADP <sup>+</sup> |
| <b>Role of chlorophyll</b>     | Captures light energy to excite electrons       | No direct role (uses ATP and NADPH produced in light reactions)                  |

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## Part 5: Connecting Cell Requirements to Photosynthesis

Autotrophs like plants meet all three cell requirements through photosynthesis and related processes:

| Cell Requirement | How Plants Meet It                                                                    | Role of Photosynthesis                                                                         |
|------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Energy</b>    | Glucose produced by photosynthesis is broken down in cellular respiration to make ATP | Photosynthesis captures light energy and stores it as chemical energy in glucose               |
| <b>Nutrients</b> | Plants absorb water (roots), CO <sub>2</sub> (leaves), and minerals (nitrates,        | CO <sub>2</sub> and H <sub>2</sub> O are direct inputs to photosynthesis; minerals are used to |



| Cell Requirement     | How Plants Meet It                                                                                                 | Role of Photosynthesis                                               |
|----------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|                      | phosphates, etc. from soil)                                                                                        | build enzymes and structures                                         |
| <b>Waste removal</b> | Oxygen (O <sub>2</sub> ) produced during photosynthesis diffuses out of cells; other wastes are stored or recycled | O <sub>2</sub> is a "waste product" of the light-dependent reactions |

## The Cyclic Nature of Life on Earth

Photosynthesis is the foundation of almost all life on Earth because it:

| Contribution                  | Explanation                                                                    |
|-------------------------------|--------------------------------------------------------------------------------|
| <b>Produces oxygen</b>        | All atmospheric oxygen comes from photosynthesis (by plants and cyanobacteria) |
| <b>Produces food</b>          | Glucose from photosynthesis feeds the entire food web (directly or indirectly) |
| <b>Removes CO<sub>2</sub></b> | Photosynthesis reduces atmospheric carbon dioxide, regulating climate          |
| <b>Stores energy</b>          | Fossil fuels (coal, oil, natural gas) are ancient photosynthetic organisms     |

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## Key Terms

| Term                             | Definition                                                                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Autotroph</b>                 | An organism that produces its own food from inorganic sources (e.g., plants, algae, cyanobacteria)                            |
| <b>Heterotroph</b>               | An organism that obtains energy and nutrients by consuming other organisms (e.g., animals, fungi)                             |
| <b>Photosynthesis</b>            | The process by which autotrophs convert light energy into chemical energy stored in glucose                                   |
| <b>Chloroplast</b>               | The organelle in plant and algal cells where photosynthesis occurs                                                            |
| <b>Chlorophyll</b>               | The green pigment in chloroplasts that captures light energy for photosynthesis                                               |
| <b>Thylakoid</b>                 | Flattened membrane sacs inside chloroplasts where light-dependent reactions occur                                             |
| <b>Grana</b>                     | Stacks of thylakoids that increase surface area for light capture                                                             |
| <b>Stroma</b>                    | The fluid-filled space inside chloroplasts where the Calvin cycle occurs                                                      |
| <b>Carotenoids</b>               | Yellow, orange, and red accessory pigments that capture additional light and provide photoprotection                          |
| <b>Light-dependent reactions</b> | The first stage of photosynthesis; requires light; produces O <sub>2</sub> , ATP, and NADPH                                   |
| <b>Calvin cycle</b>              | The second stage of photosynthesis; does not require light directly; uses CO <sub>2</sub> , ATP, and NADPH to produce glucose |
| <b>ATP</b>                       | Adenosine triphosphate; the energy currency of the cell                                                                       |

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## Review Questions

*Responses should be written on a separate sheet of paper.*

1. List the three universal requirements of all cells (energy, nutrients, waste removal). For each requirement, explain why cells need it and give one example of a specific substance involved.
2. Identify where photosynthesis takes place within the cells of autotrophs. Your answer must include the name of the organelle, the type of cell that contains this organelle (plant, animal, or both), and which organisms are autotrophs.
3. Describe the role of chlorophyll and other pigments in capturing light energy. Your answer must include what chlorophyll does when light strikes it, the difference between chlorophyll a and accessory pigments (chlorophyll b and carotenoids), and why plants have multiple pigments rather than just one.
4. Explain why plants appear green to the human eye. Use the terms "absorb," "reflect," and "wavelength" in your answer.
5. Application question: A student places a green plant in a sealed glass container with no light. The plant has water and soil. Answer the following: Will the plant perform photosynthesis? Why or why not? What will happen to the plant over time, and which cell requirement is not being met?
6. Thinking question: If all autotrophs (plants, algae, cyanobacteria) suddenly disappeared from Earth, explain what would happen to oxygen levels in the atmosphere, carbon dioxide levels, and heterotrophs (including humans).